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Computer Vision

Computer Vision (S2-25_AIMLZG525)

Session #3:
Low Level Vision

Acknowledgement: Slide Materials adopted from - Intro to Computer Vision (Cornell Tech); Noah Snavely

Raghavendra Singh



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Topics

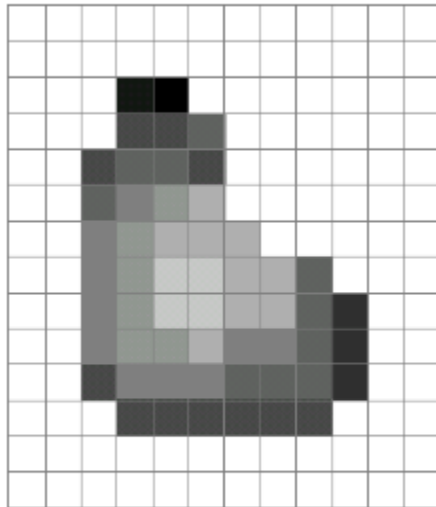
- Image Histogram & Histogram Equalization
- Basic Intensity Transformations
- Image Smoothing

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What is an image?

A grid (matrix) of intensity values



=

255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	20	0	255	255	255	255	255	255	255
255	255	255	75	75	75	255	255	255	255	255	255
255	255	75	95	95	75	255	255	255	255	255	255
255	255	96	127	145	175	255	255	255	255	255	255
255	255	127	145	175	175	175	255	255	255	255	255
255	255	127	145	200	200	175	175	95	255	255	255
255	255	127	145	200	200	175	175	95	47	255	255
255	255	127	145	145	175	127	127	95	47	255	255
255	255	74	127	127	127	95	95	95	47	255	255
255	255	255	74	74	74	74	74	74	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255

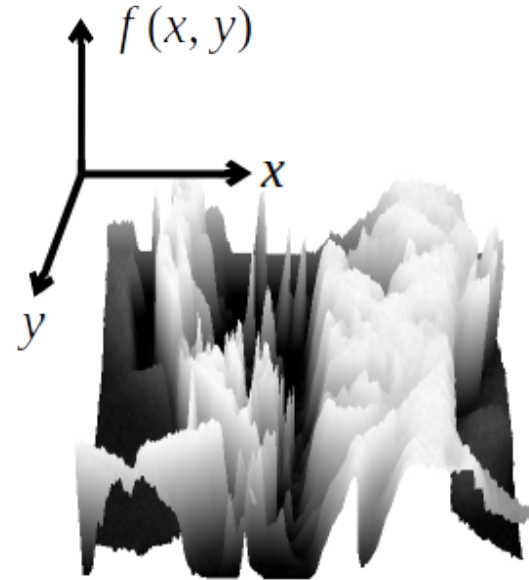


What is an image?

Can think of a (grayscale) image as a function f from $\mathbb{R}^2$ to $\mathbb{R}$:
 $f(x,y)$ gives the intensity at position (x,y)



[snoop](#)



[3D view](#)

Image Transformations

As with any function, we can apply operators to an image



$$g(x,y) = f(x,y) + 20$$



$$g(x,y) = f(-x,y)$$



Image Histogram

Un-normalized histogram: $h(r_k) = n_k$ for $k = 0, 1, 2, \dots, L - 1$

Note: n_k is the number of pixels in f with intensity r_k

Normalized histogram: $p(r_k) = \frac{h(r_k)}{MN} = \frac{n_k}{MN}$

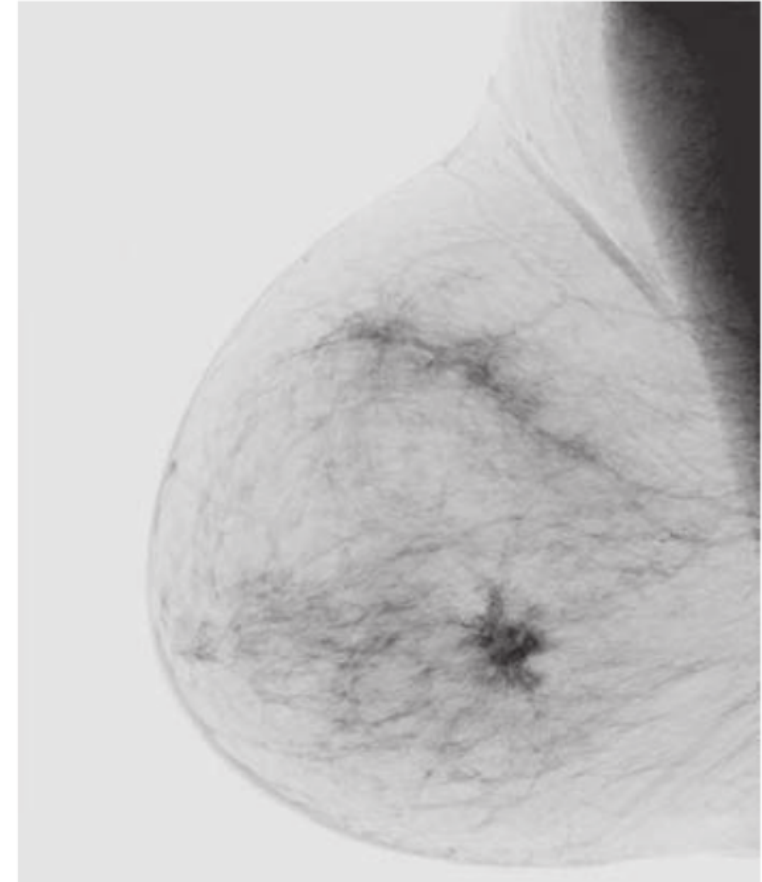
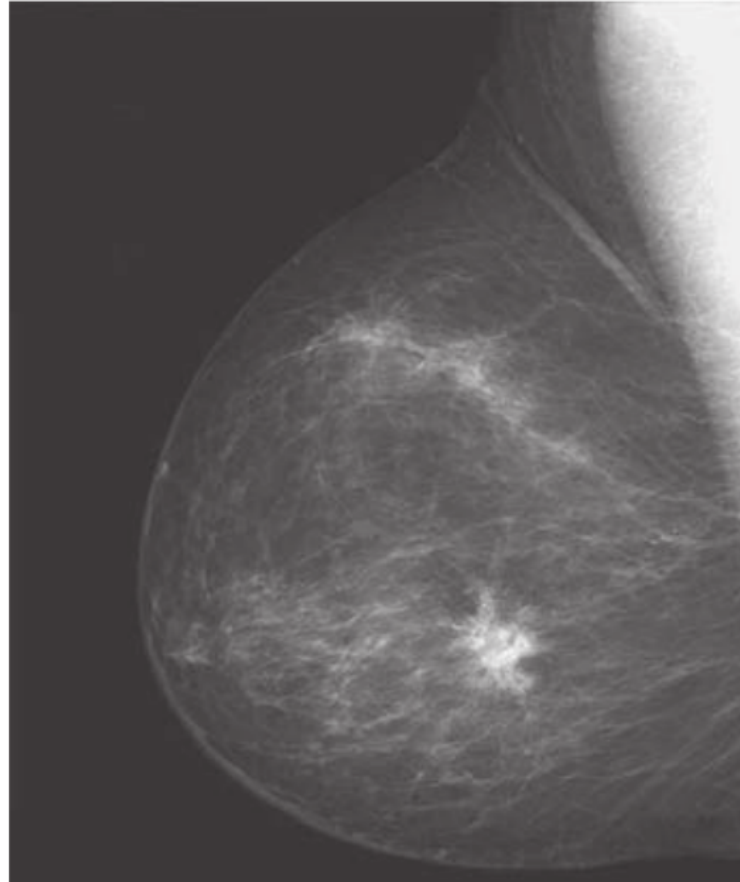
Note: M and N are the number of image rows and columns

The sum of $p(r_k)$ for all values of k is always 1

Mostly, we work with normalized histograms

Elementary Intensity Transformations

$$s = L - 1 - r$$



A digital mammogram and its Negative image



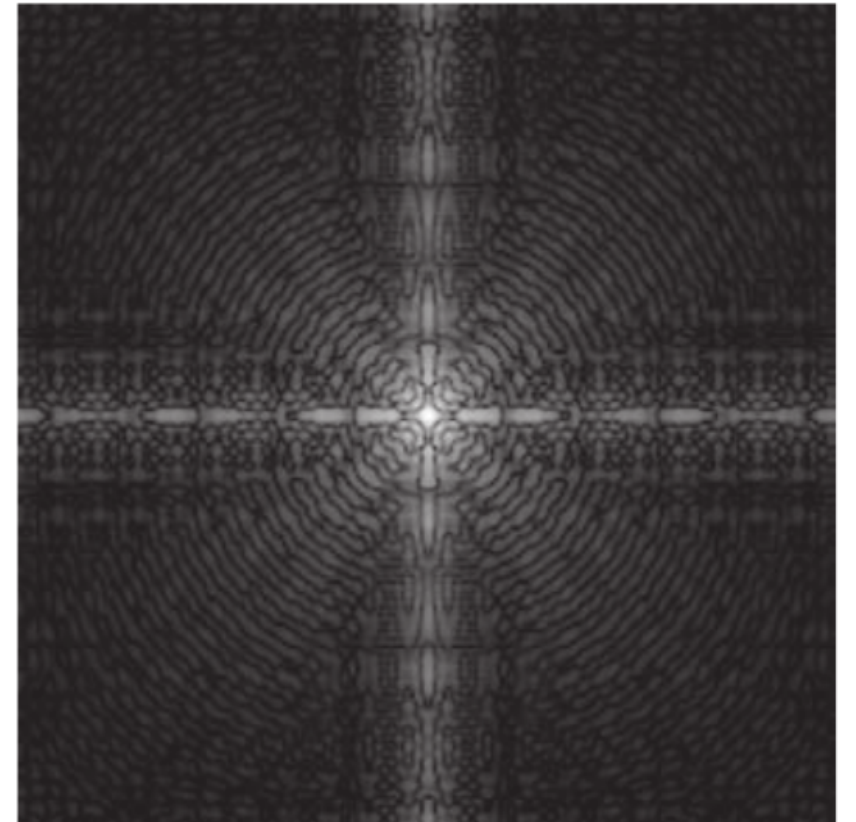
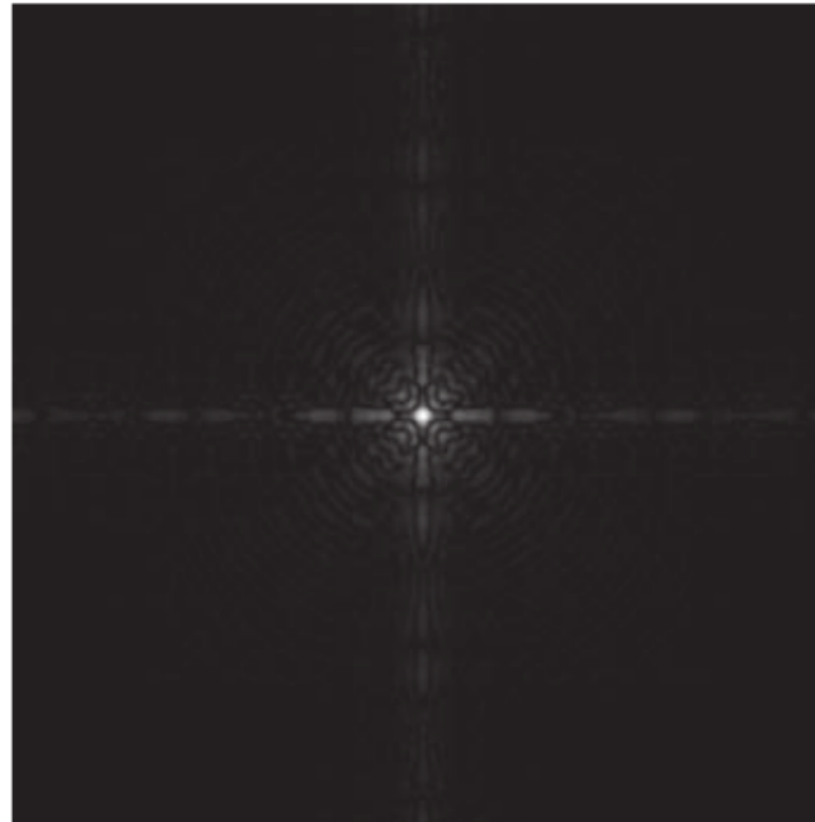
Elementary Intensity Transformations

Log Transformations

$$s = c \log(1 + r)$$

Fourier spectrum displayed as a grayscale image and the result of applying the log transformation in with $c = 1$.

Both images are scaled to the range $[0, 255]$.





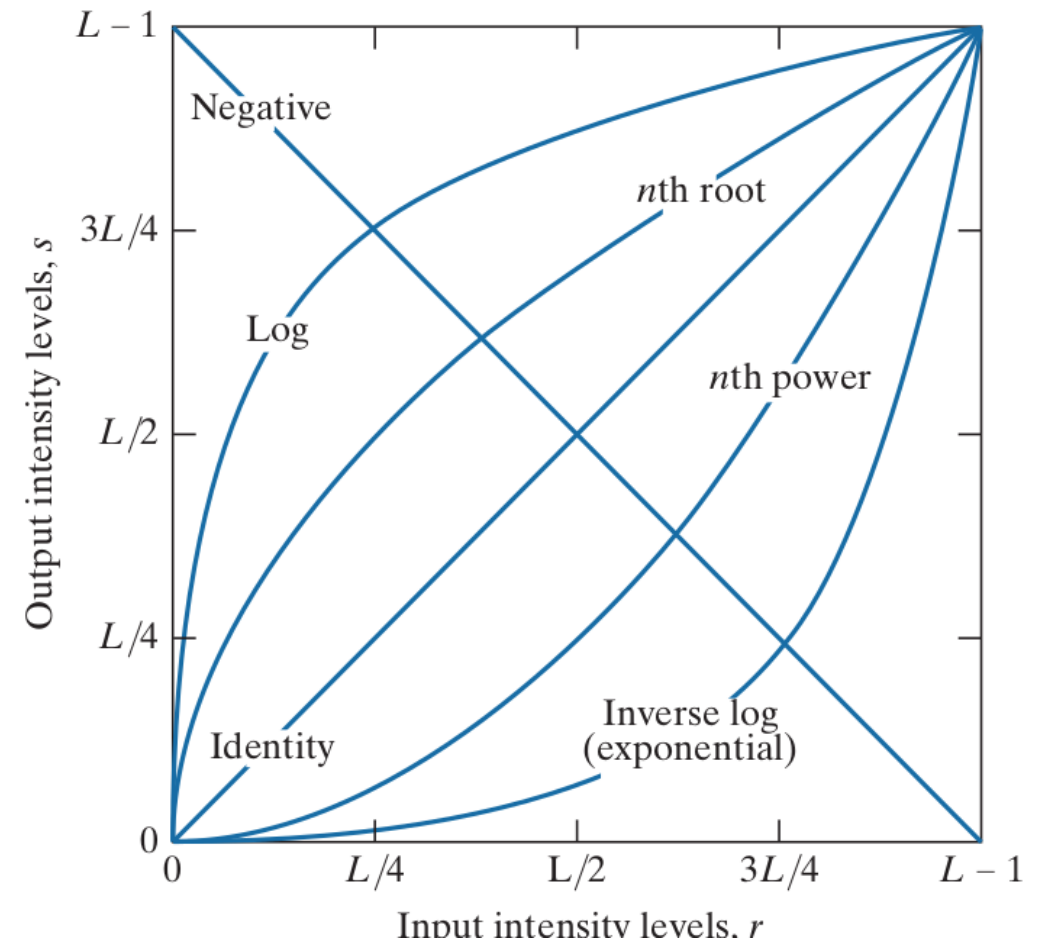
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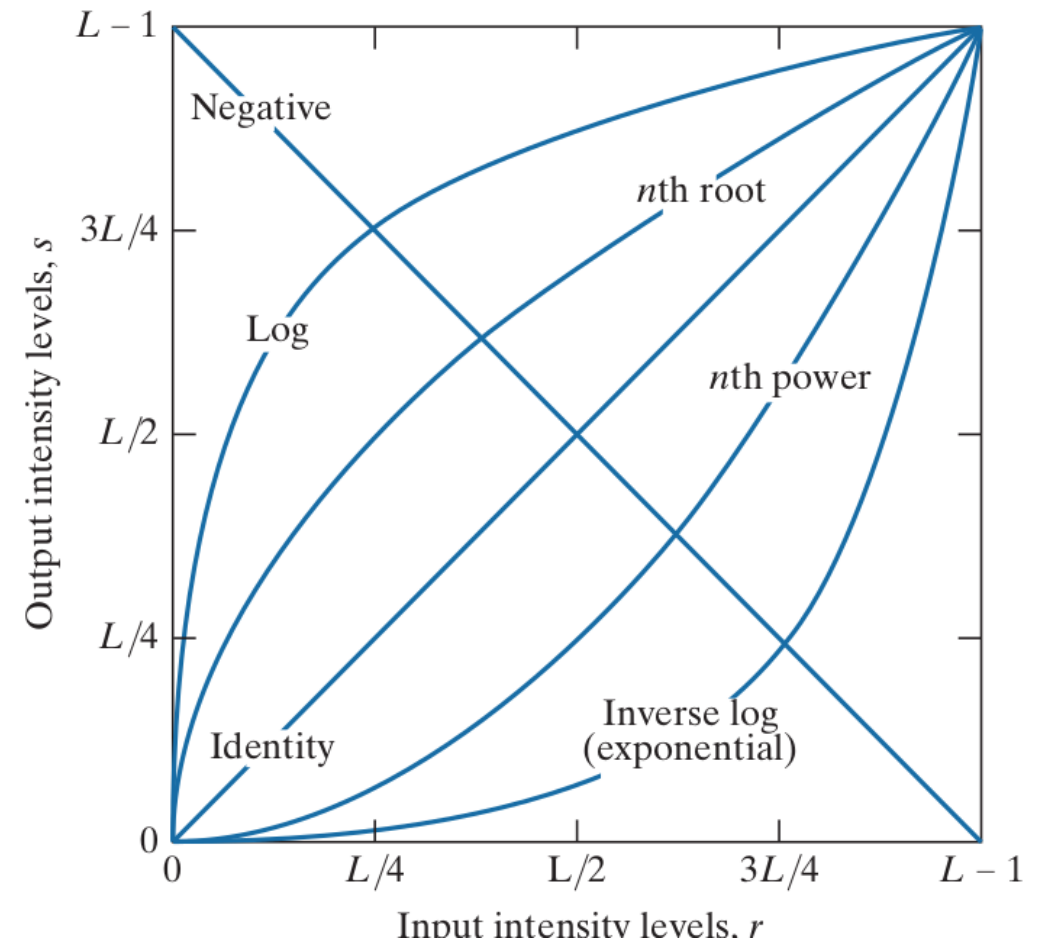


Elementary Intensity Transformations

Power-Law Transformations

$$s = cr^\gamma$$

- c and γ are positive constants





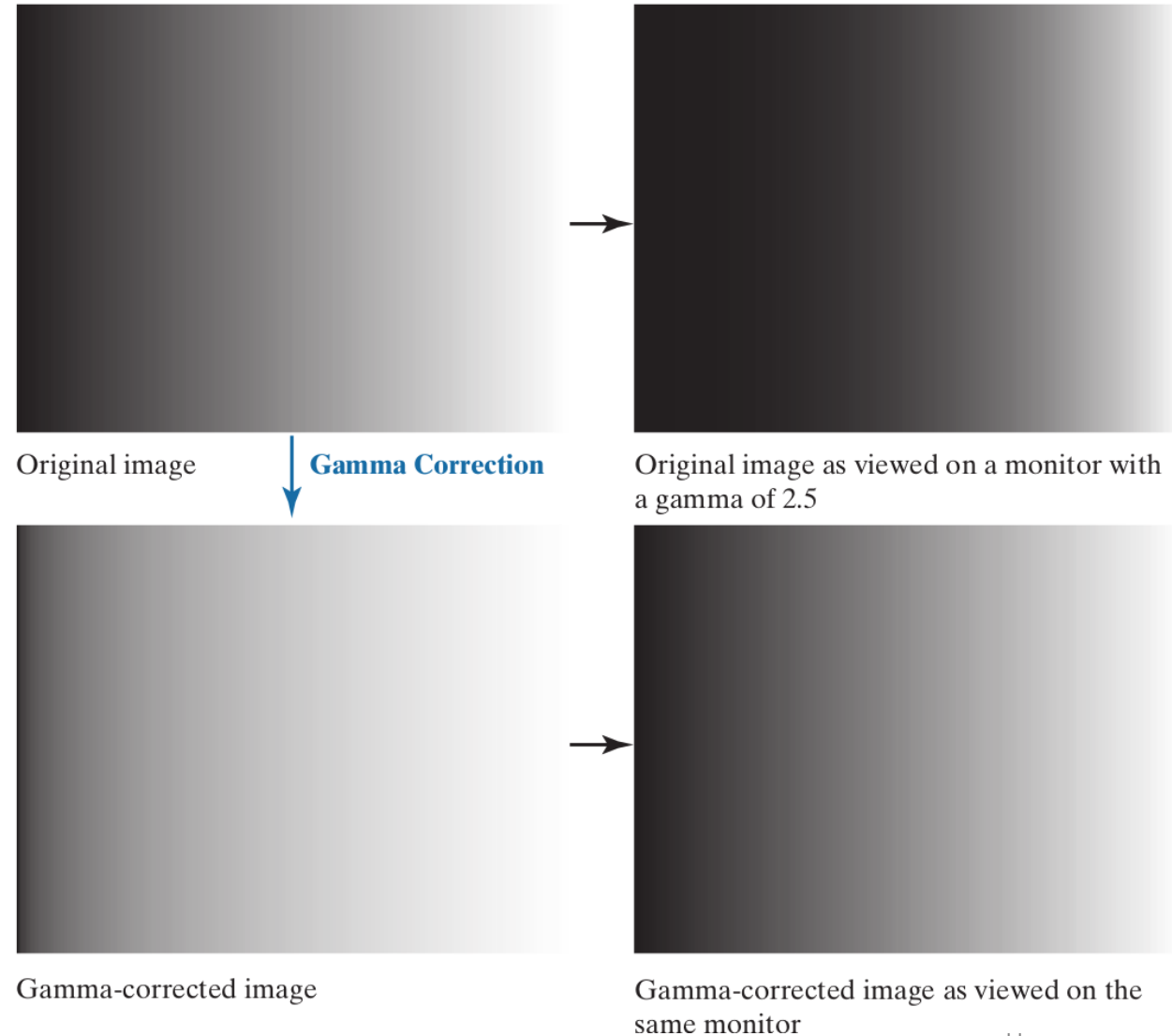
Elementary Intensity Transformations

Power-Law Transformations- Applications

Gamma correction

- Cathode ray tube (CRT) devices have an intensity-to-voltage response that is a power function, with exponents varying from approximately 1.8 to 2.5.

$$S = cr^\gamma$$





Elementary Intensity Transformations

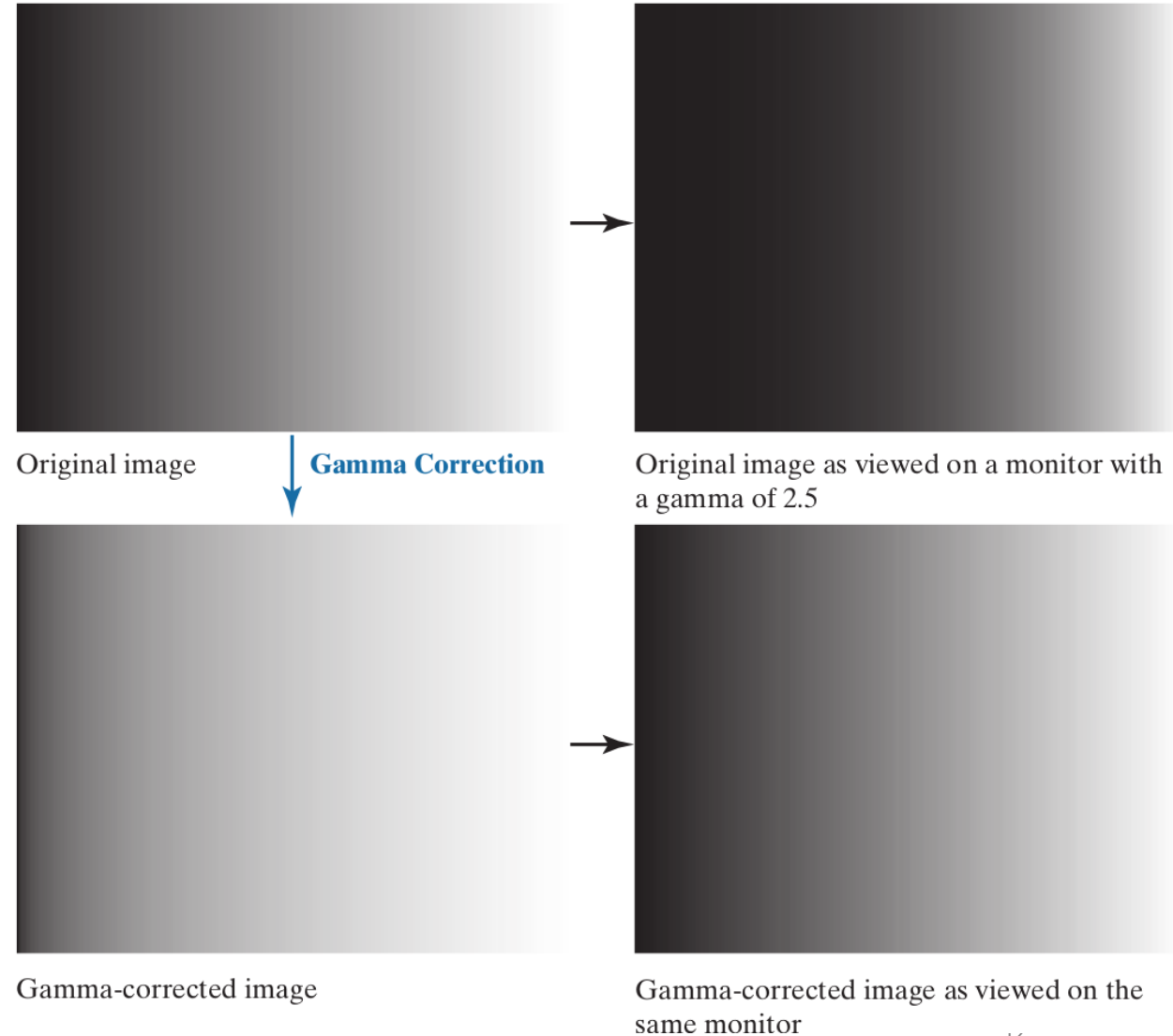
Power-Law Transformations- Applications

Gamma correction

- Cathode ray tube (CRT) devices have an intensity-to-voltage response that is a power function, with exponents varying from approximately 1.8 to 2.5.
- Gamma correction consists of using the transformation

$$s = r^{1/2.5} = r^{0.4}$$

$$s = cr^\gamma$$



Elementary Intensity Transformations

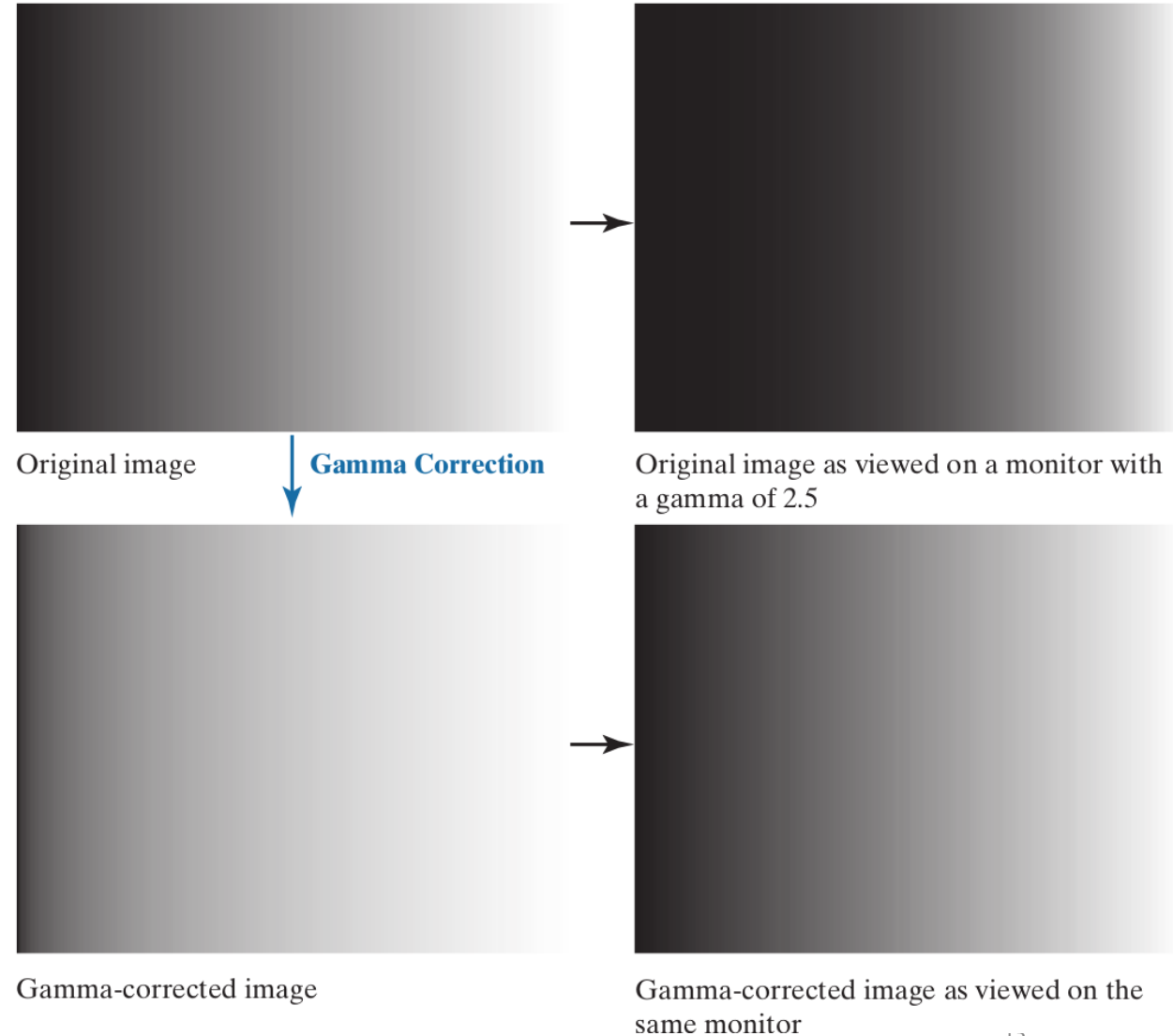
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Elementary Intensity Transformations

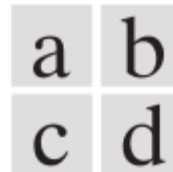
Power-Law Transformations- Applications

Medical Imaging

(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the transformation with $c = 1$ and $g = 0.6, 0.4$, and 0.3 , respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

$$s = cr^\gamma$$





Elementary Intensity Transformations

Power-Law Transformations- Applications

Aerial Imaging

(a) Aerial image.

(b)–(d) Results of applying the transformation in with $\gamma = 3.0$, 4.0 , and 5.0 , respectively. ($c = 1$ in all cases.)

(Original image, courtesy of NASA.)



a	b
c	d

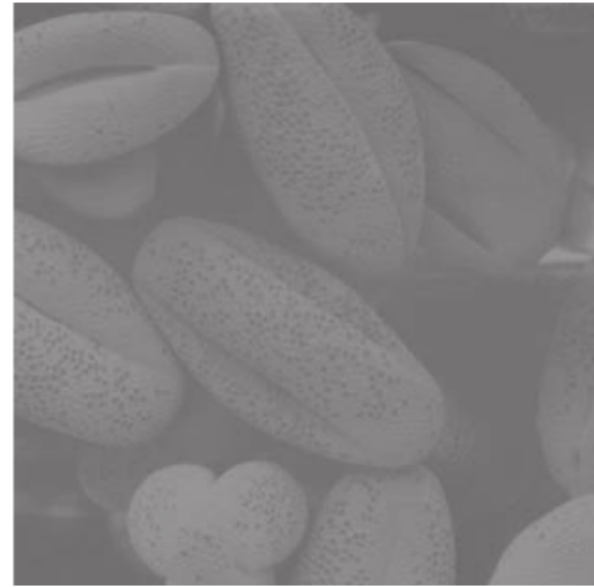
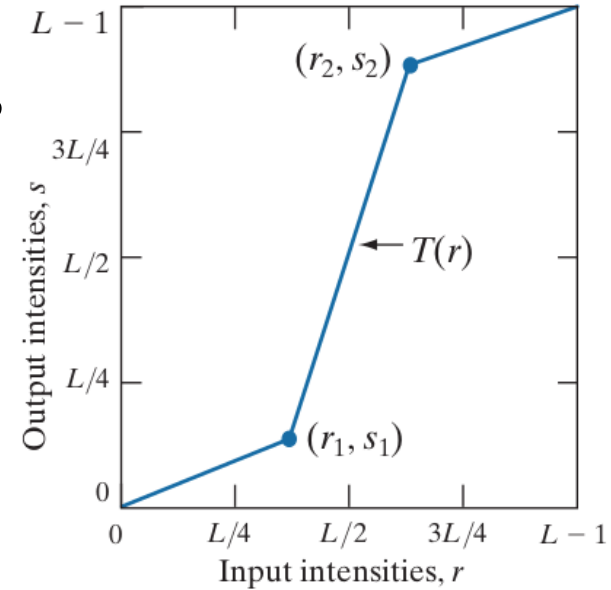
$$s = cr^\gamma$$



Elementary Intensity Transformations

Contrast Stretching; Thresholding

- (a) Piecewise linear transformation function.
- (b) A low-contrast electron microscope image of pollen, magnified 700 times.
- (c) Result of contrast stretching.
- (d) Result of thresholding.

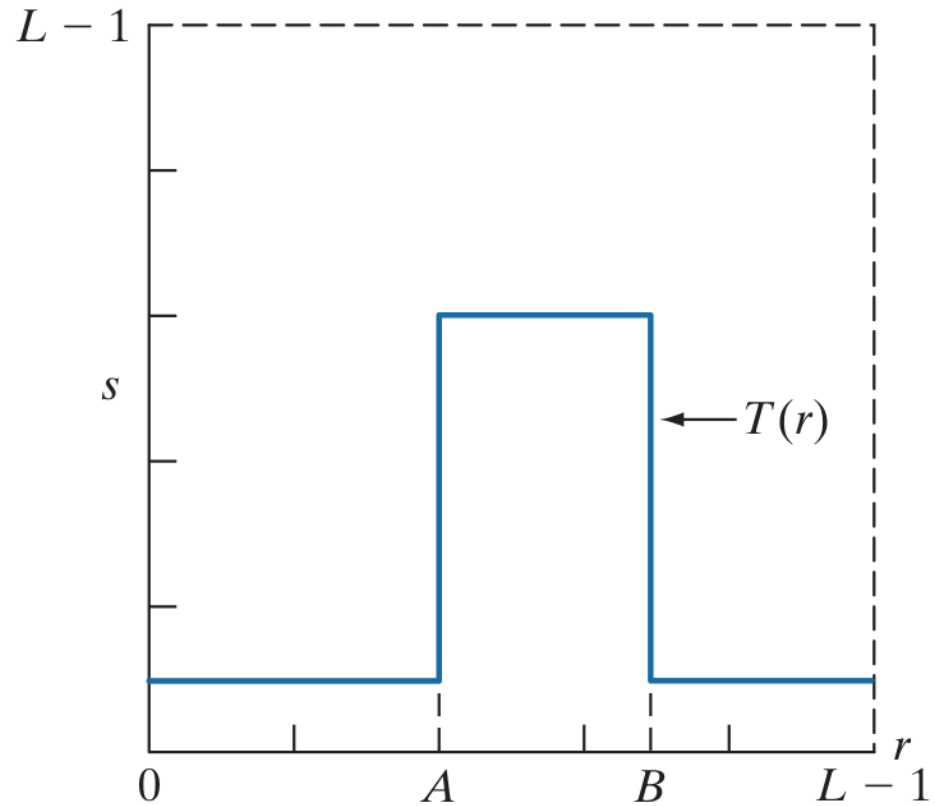


a	b
c	d

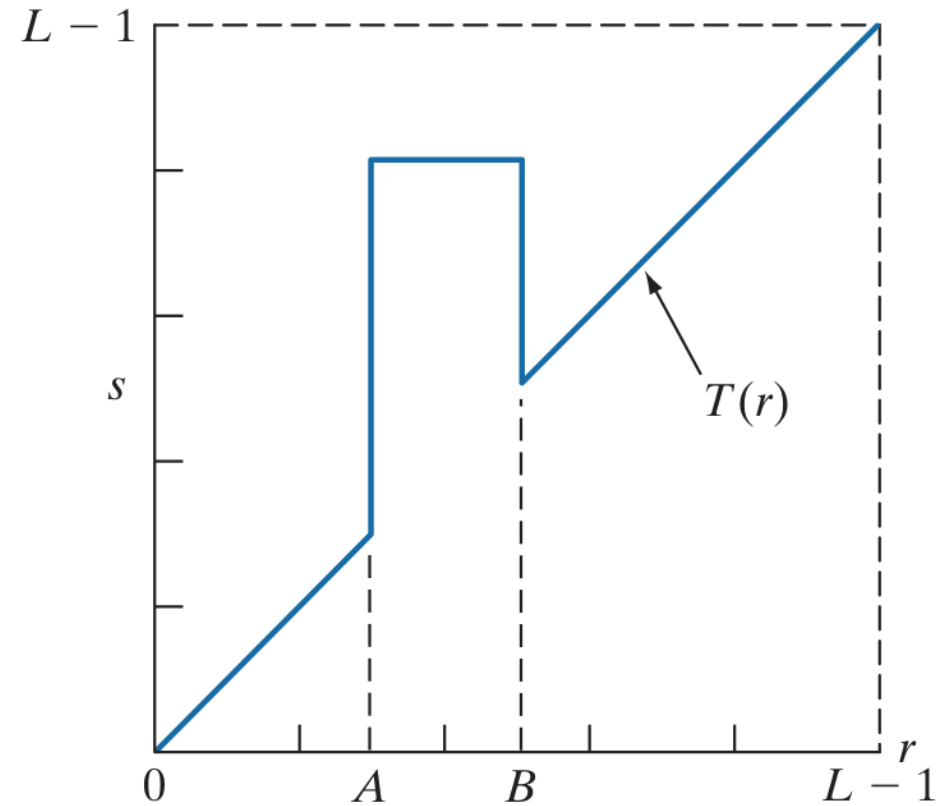


Elementary Intensity Transformations

Intensity Level Slicing

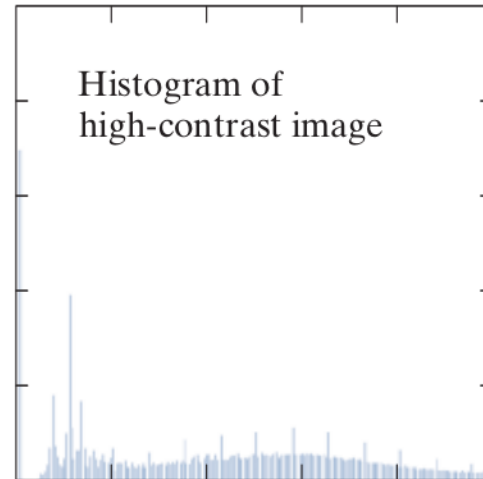
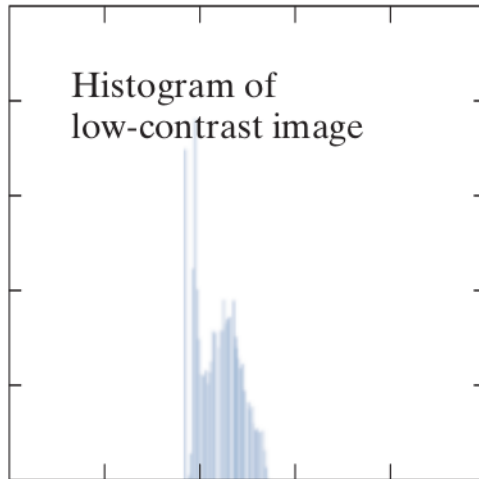
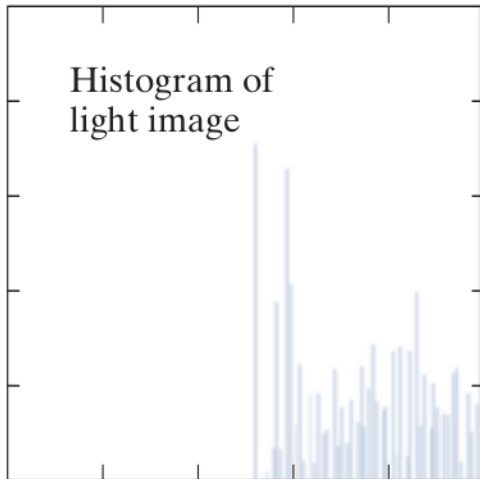
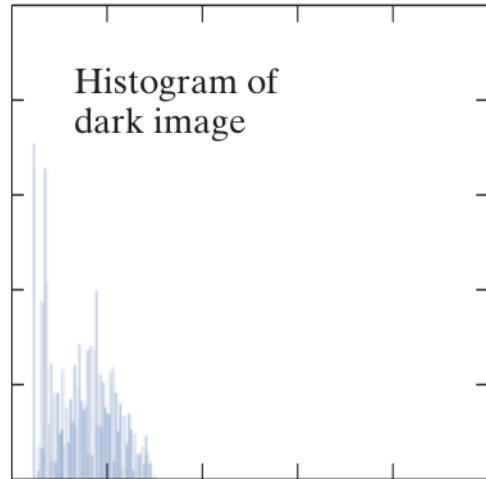
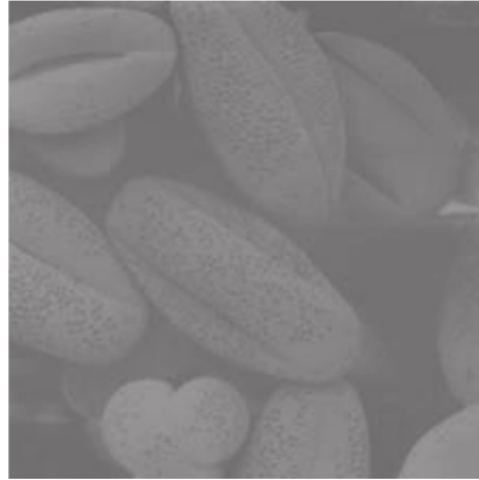
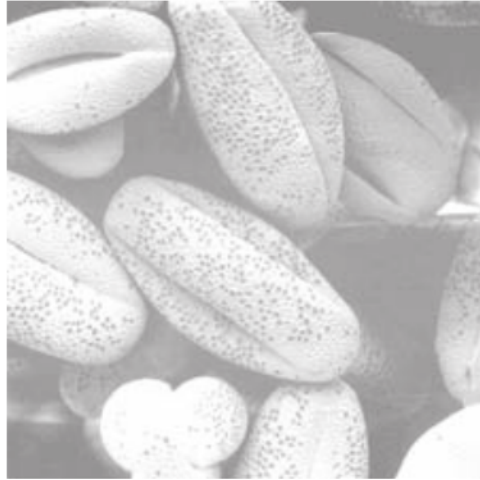


Highlights range $[A, B]$ and reduces all other intensities to a lower level.



This function highlights range $[A, B]$ and leaves other intensities unchanged.

Image Histogram



Histogram shape is related to image appearance ;



Histogram Equalization

Histogram :

$$p(r) = \left(\frac{\text{Number of pixels with intensity } r}{\text{Total number of pixels}} \right)$$

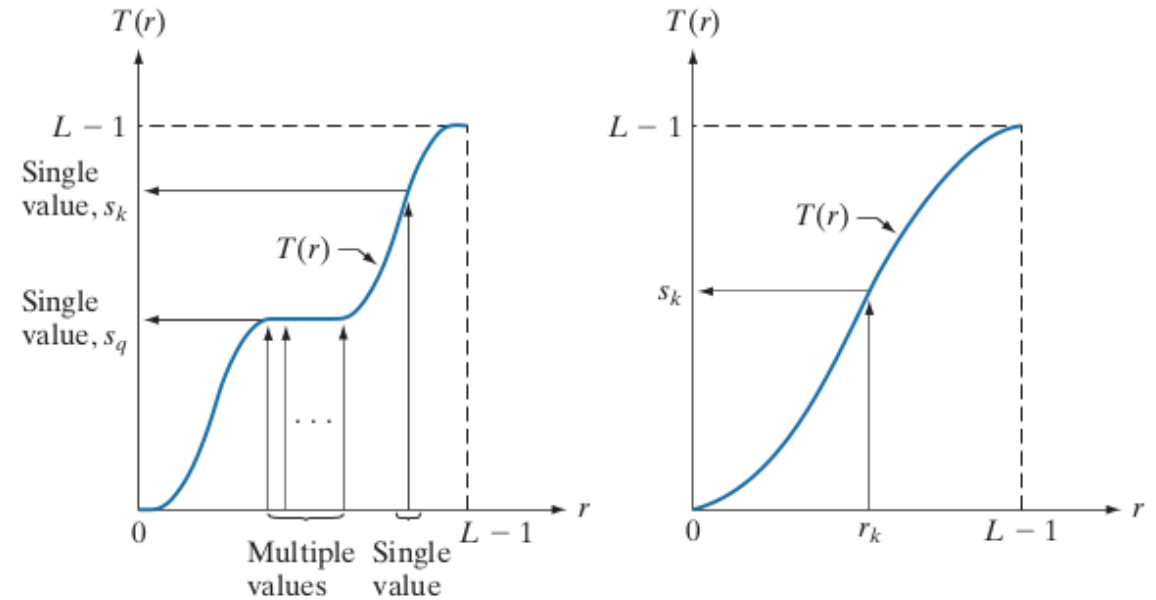
To do :

Intensity mapping

$$s = T(r)$$

Assume

- $T(r)$ is single-valued and monotonically increasing.
- $0 \leq T(r) \leq 1$ and $0 \leq r \leq 1$



[†] A function $T(r)$ is a *monotonic increasing* function if $T(r_2) \geq T(r_1)$ for $r_2 > r_1$. $T(r)$ is a *strictly monotonic increasing* function if $T(r_2) > T(r_1)$ for $r_2 > r_1$. Similar definitions apply to a monotonic decreasing function.

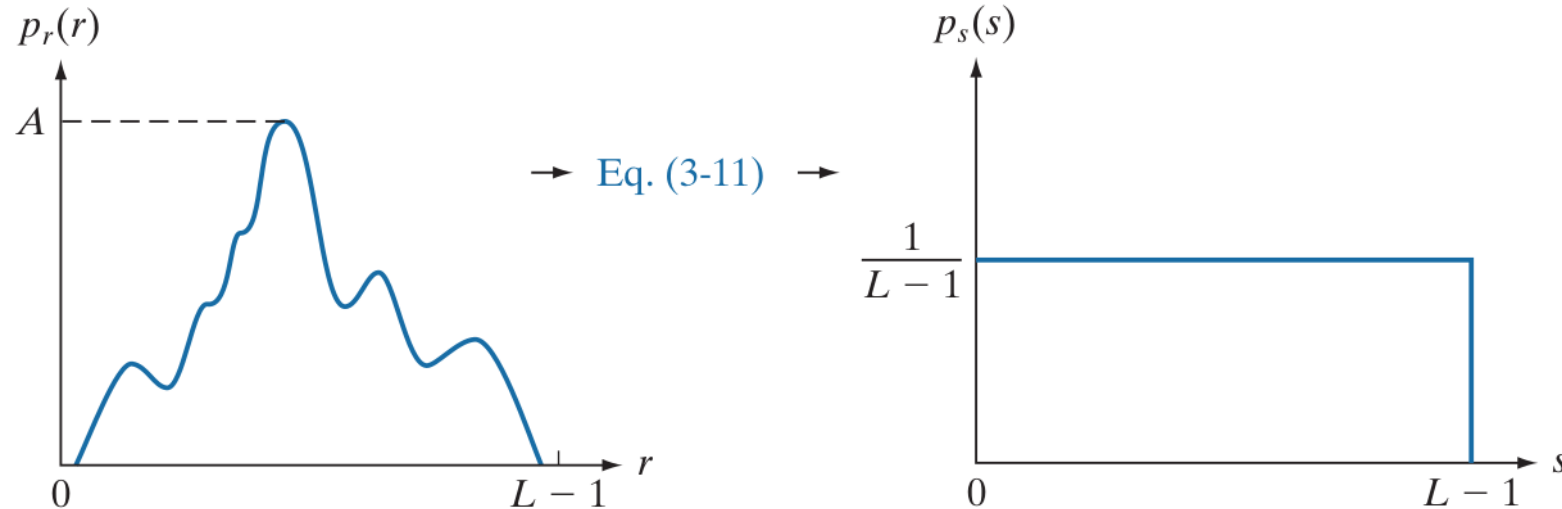


Histogram Equalization

Histogram :

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Histogram Equalization

$$s = T(r) \quad 0 \leq r \leq L - 1$$

- a. $T(r)$ is a strictly monotonically increasing function in the interval $0 \leq r \leq L - 1$;
- b. $0 \leq T(r) \leq L - 1$ for $0 \leq r \leq L - 1$.

If $p_r(r)$ and $T(r)$ are known, and

$T(r)$ is continuous and differentiable over the range of values of interest, then the PDF of the transformed (mapped) variable s can be obtained as

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$



Histogram Equalization

If $p_r(r)$ and $T(r)$ are known, and

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$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$

A transformation function of particular importance in image processing is

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$



Histogram Equalization

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$

trying to compute

$$\frac{ds}{dr} = \frac{dT(r)}{dr}$$

$$= (L-1) \frac{d}{dr} \left[\int_0^r p_r(w) dw \right]$$

$$= (L-1) p_r(r) \quad \text{Leibniz's rule in calculus}$$

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$

$$= p_r(r) \left| \frac{1}{(L-1)p_r(r)} \right|$$

$$= \frac{1}{L-1} \quad 0 \leq s \leq L-1$$

Form of uniform probability density function



Histogram Equalization

T(r) for continuous values:

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

Digital image is discrete:

$$p_r(r_k) = \frac{n_k}{MN}$$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j) \quad k = 0, 1, 2, \dots, L - 1$$

Need to just use this to transform image



Histogram Equalization - Example

Suppose that a 3-bit image ($L=8$) of size 64×64 pixels ($MN = 4096$) has the intensity distribution shown in following table.

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02



Histogram Equalization - Example

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
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$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$s_0 = T(r_0) = 7 \sum_{j=0}^0 p_r(r_j) = 7 \times 0.19 = 1.33 \rightarrow 1$$

$$s_1 = T(r_1) = 7 \sum_{j=0}^1 p_r(r_j) = 7 \times (0.19 + 0.25) = 3.08 \rightarrow 3$$

$$s_2 = 4.55 \rightarrow 5 \quad s_3 = 5.67 \rightarrow 6$$

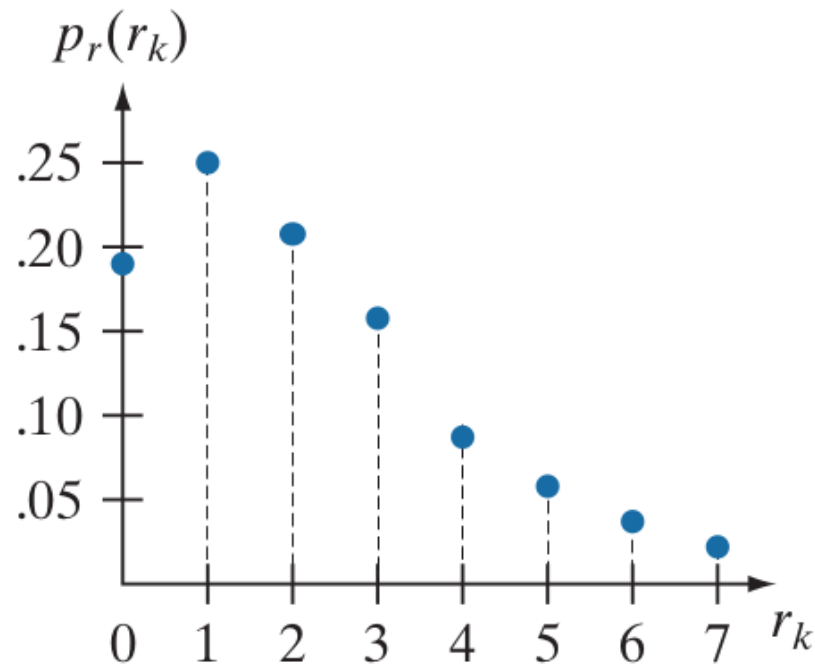
$$s_4 = 6.23 \rightarrow 6 \quad s_5 = 6.65 \rightarrow 7$$

$$s_6 = 6.86 \rightarrow 7 \quad s_7 = 7.00 \rightarrow 7$$



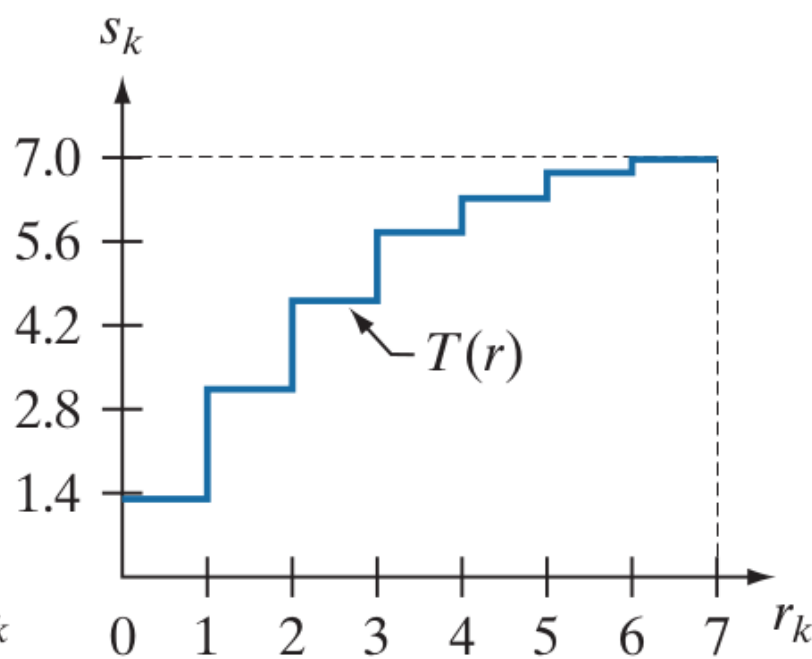
Histogram Equalization - Example

$$\begin{array}{llll} s_0 = 1.33 \rightarrow 1 & s_2 = 4.55 \rightarrow 5 & s_4 = 6.23 \rightarrow 6 & s_6 = 6.86 \rightarrow 7 \\ s_1 = 3.08 \rightarrow 3 & s_3 = 5.67 \rightarrow 6 & s_5 = 6.65 \rightarrow 7 & s_7 = 7.00 \rightarrow 7 \end{array}$$

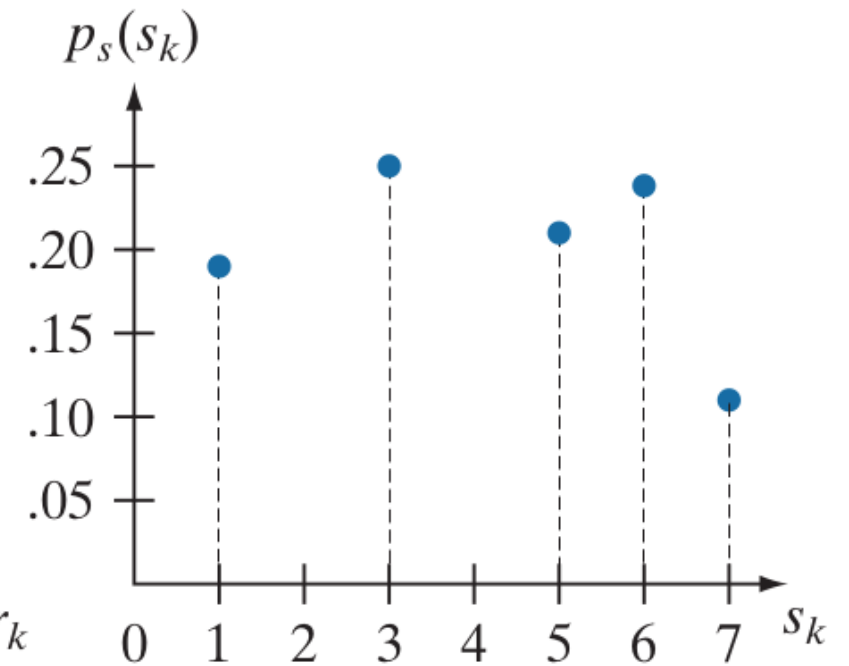


Original histogram.

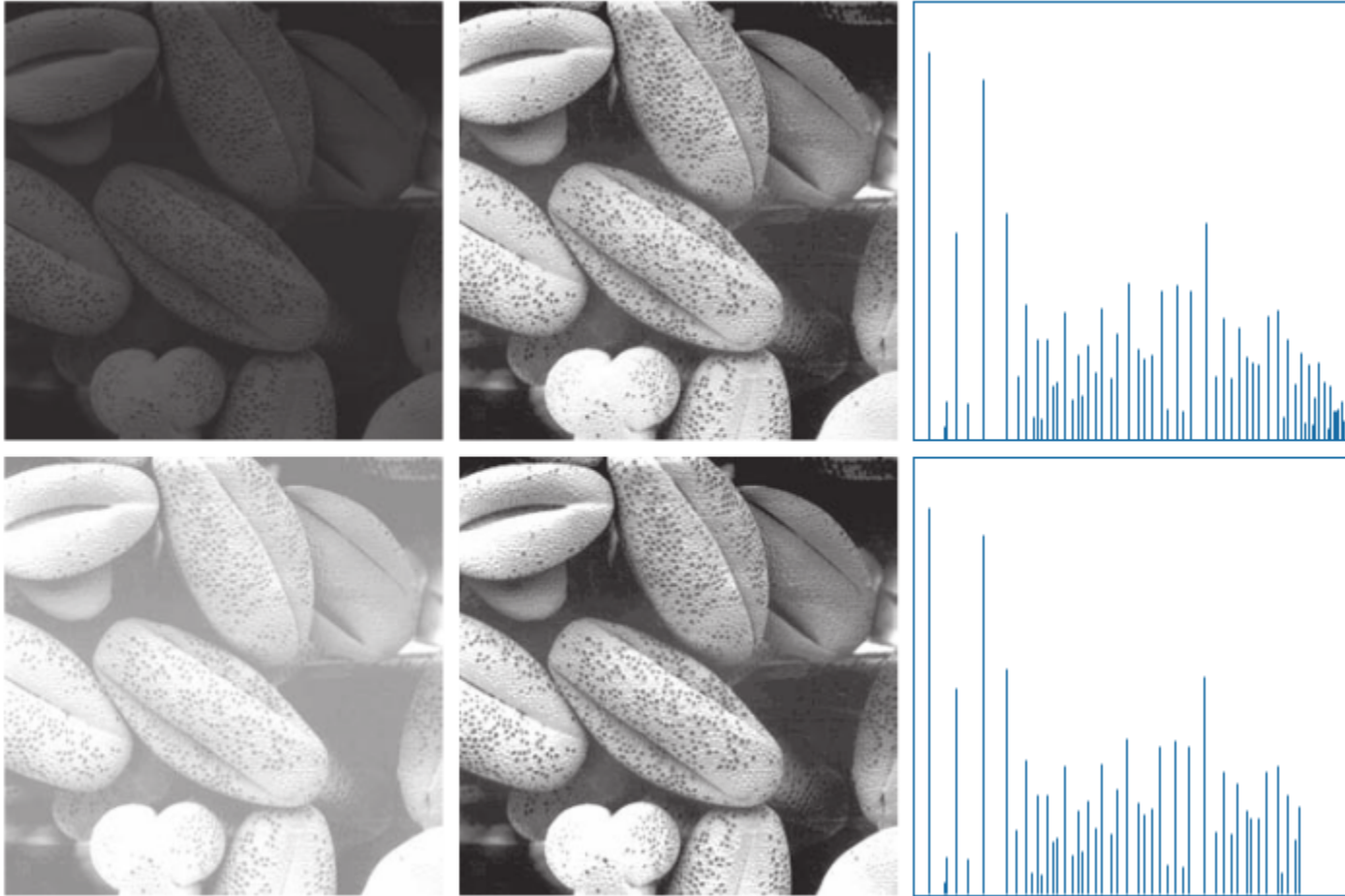
Equalized histogram.



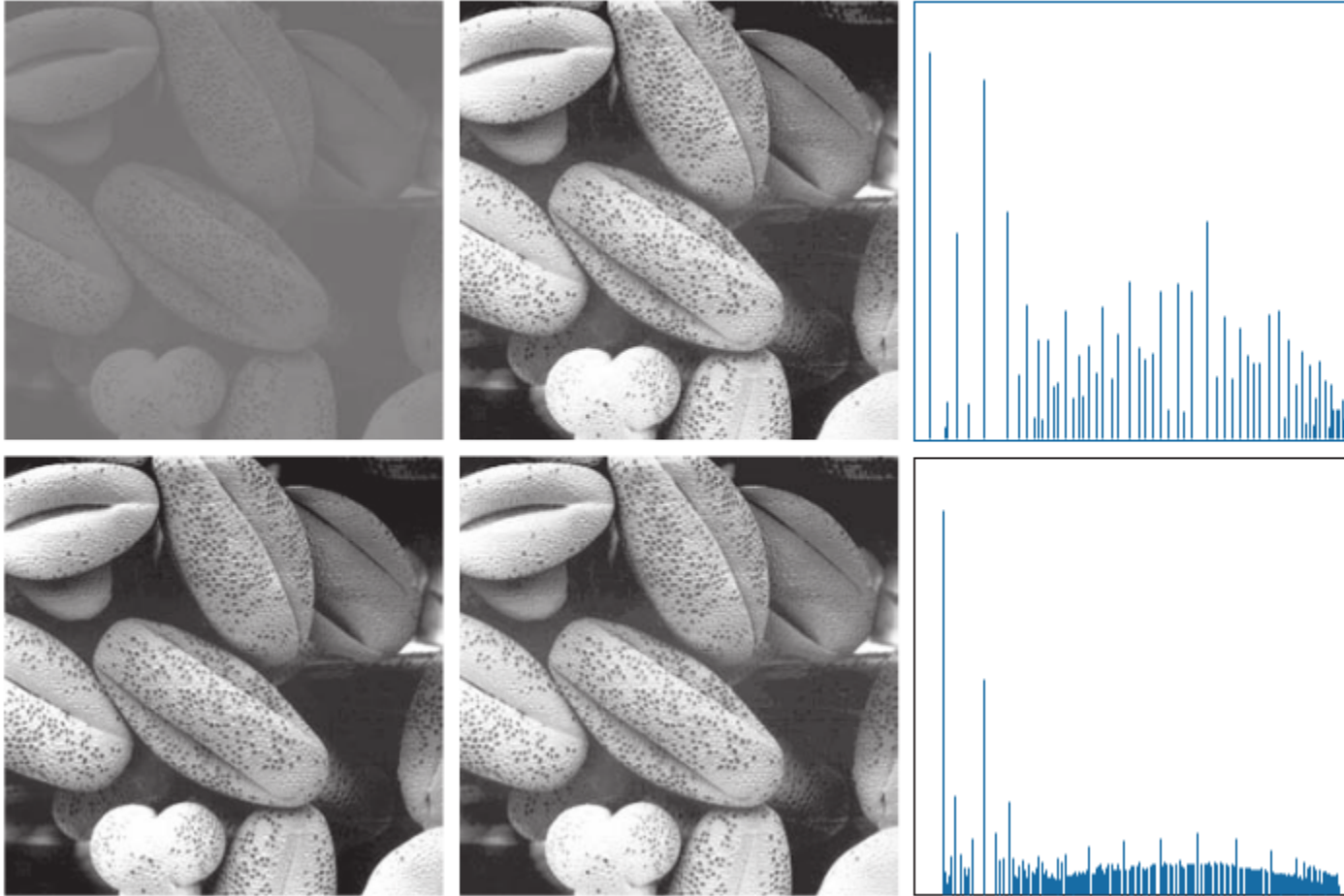
Transformation function.



Histogram Equalization - Example



Histogram Equalization - Example



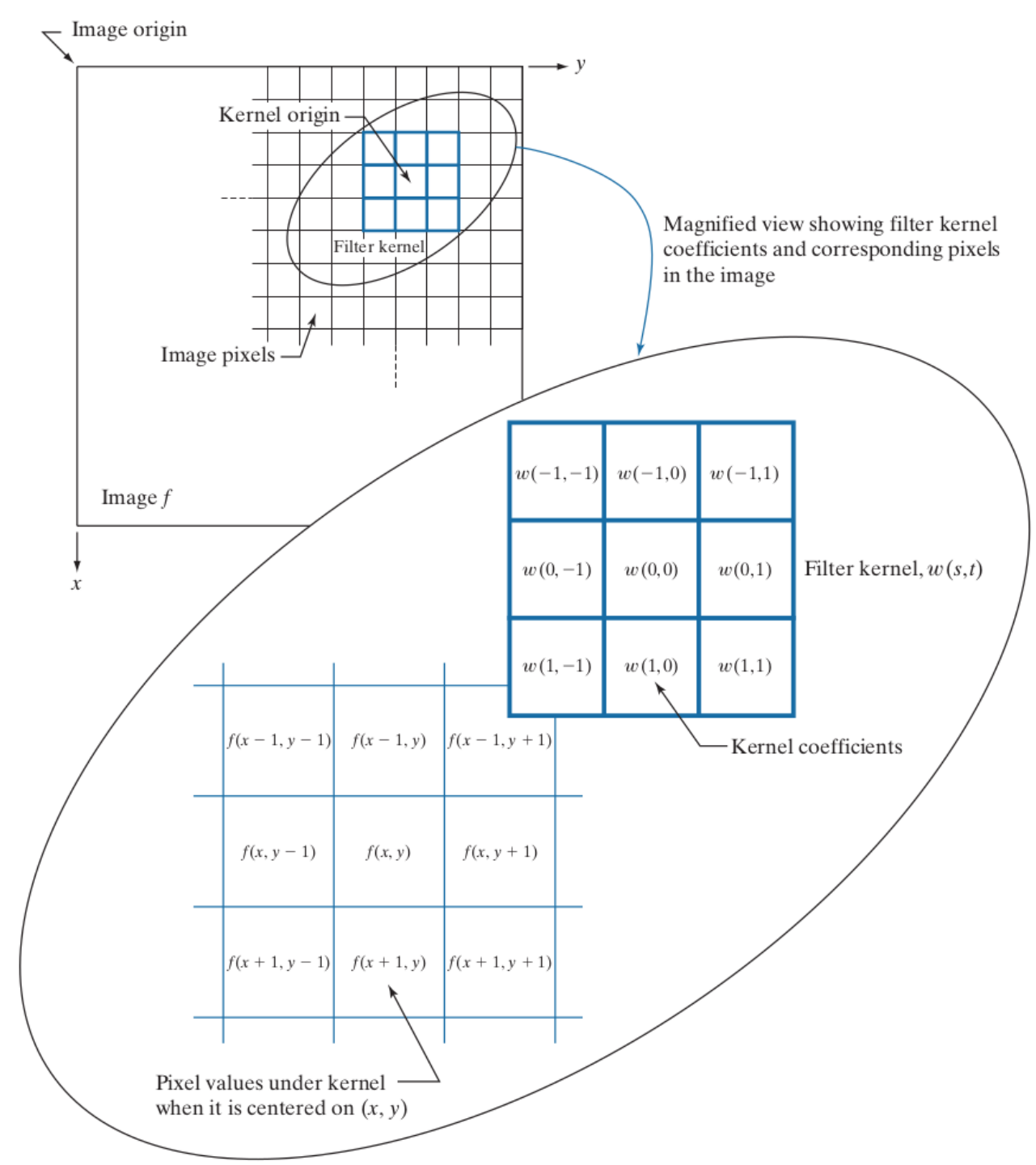


Spatial Filtering

Note:

Origin of the image is at the top left, but the origin of the kernel is at its center.

Placing the origin at the center of spatially symmetric kernels simplifies writing expressions for linear filtering.





Spatial Filtering

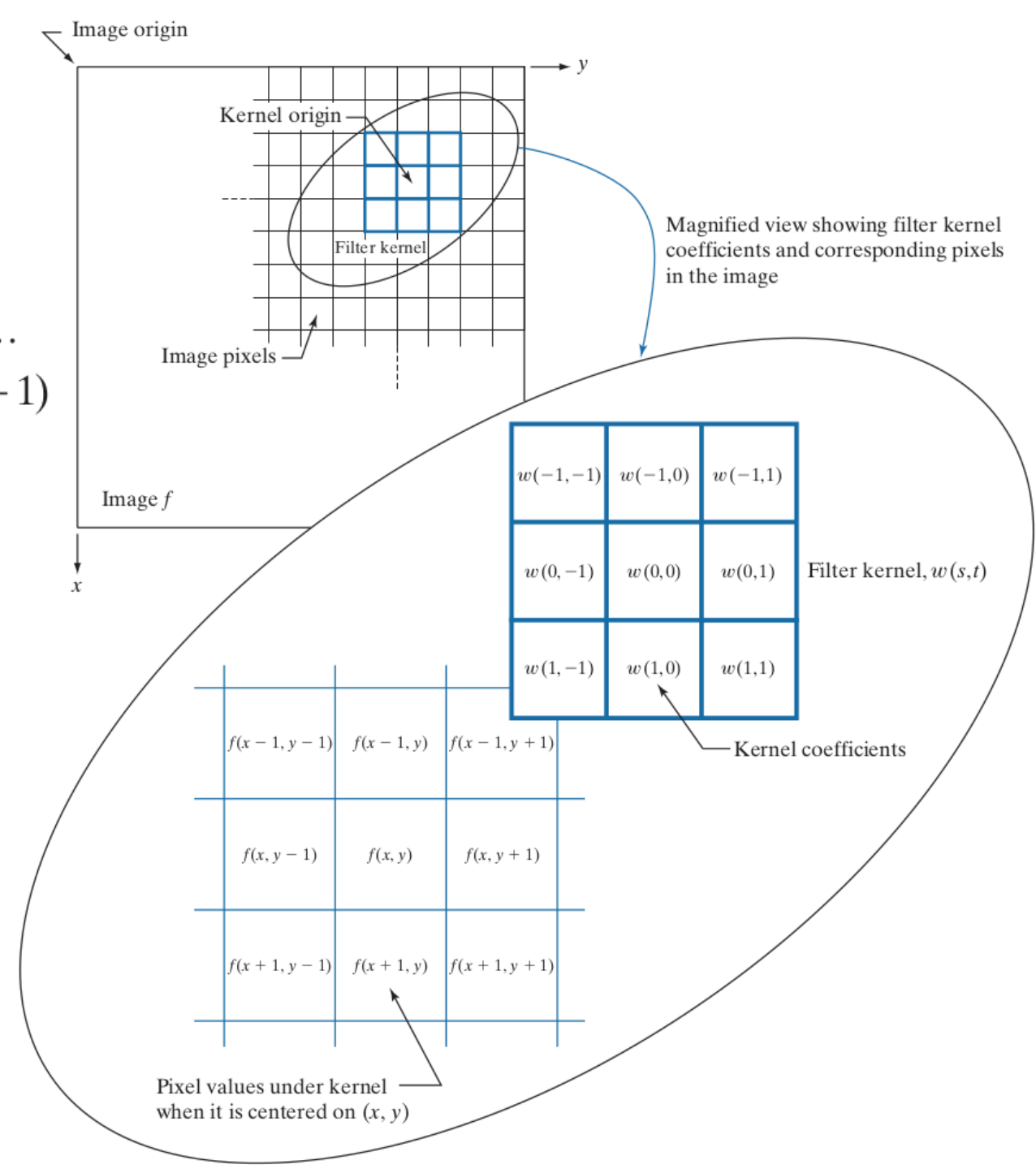
$$g(x, y) = w(-1, -1)f(x - 1, y - 1) + w(-1, 0)f(x - 1, y) + \dots \\ + w(0, 0)f(x, y) + \dots + w(1, 1)f(x + 1, y + 1)$$

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t)f(x + s, y + t)$$

Note:

$$m = 2a + 1$$

$$n = 2b + 1$$





Spatial Filtering

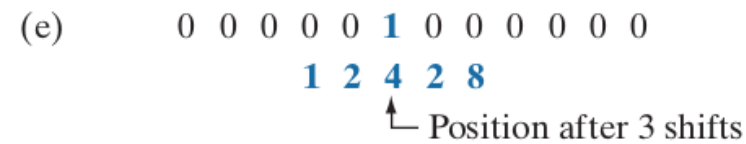
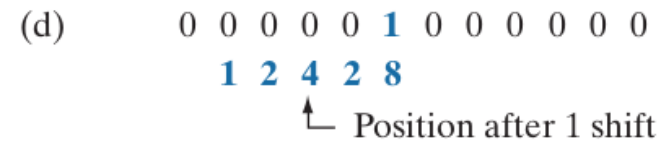
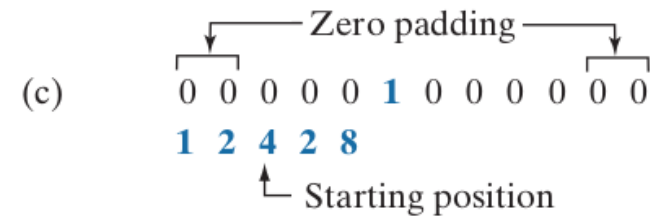
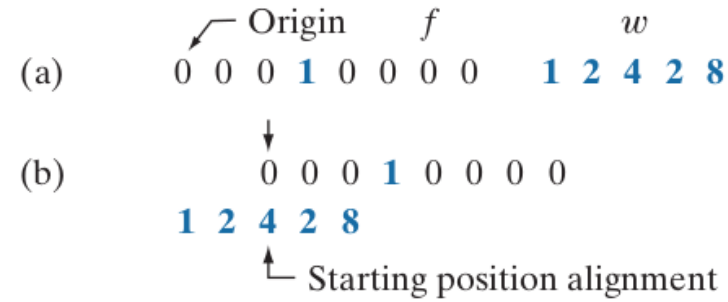
- A spatial filter consists of (a) **a neighborhood**, and (b) **a predefined operation**
- Linear spatial filtering of an image of size MxN with a filter of size mxn is given by the expression

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$



Spatial Filtering

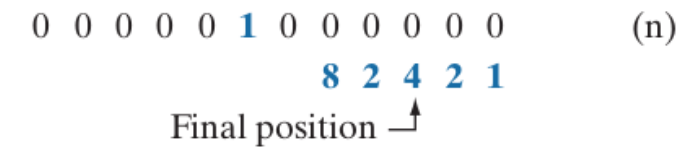
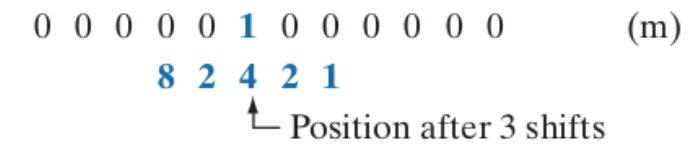
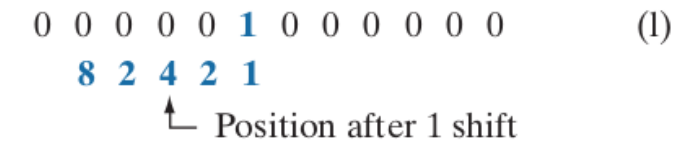
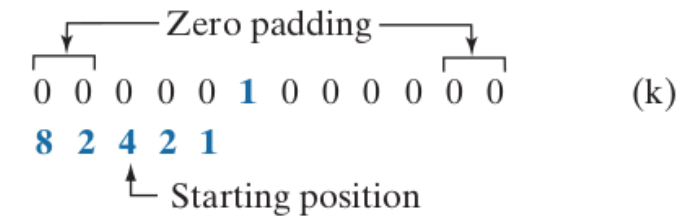
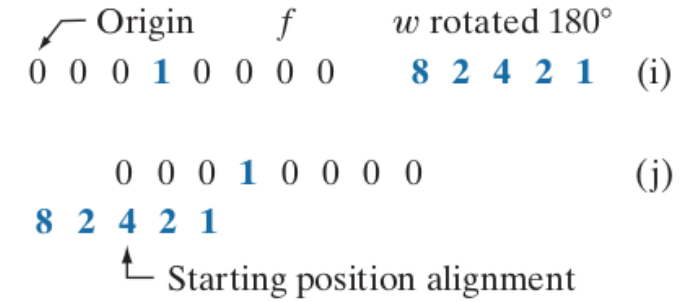
Correlation



Correlation result

(g) 0 8 2 4 2 1 0 0

Convolution



Convolution result

(o) 0 1 2 4 2 8 0 0

(a)

(c)

(f)



Smoothing Spatial Filters

$$\frac{1}{9} \times$$

1	1	1
1	1	1
1	1	1

Box kernel

Gaussian kernel.

$$\frac{1}{4.8976} \times$$

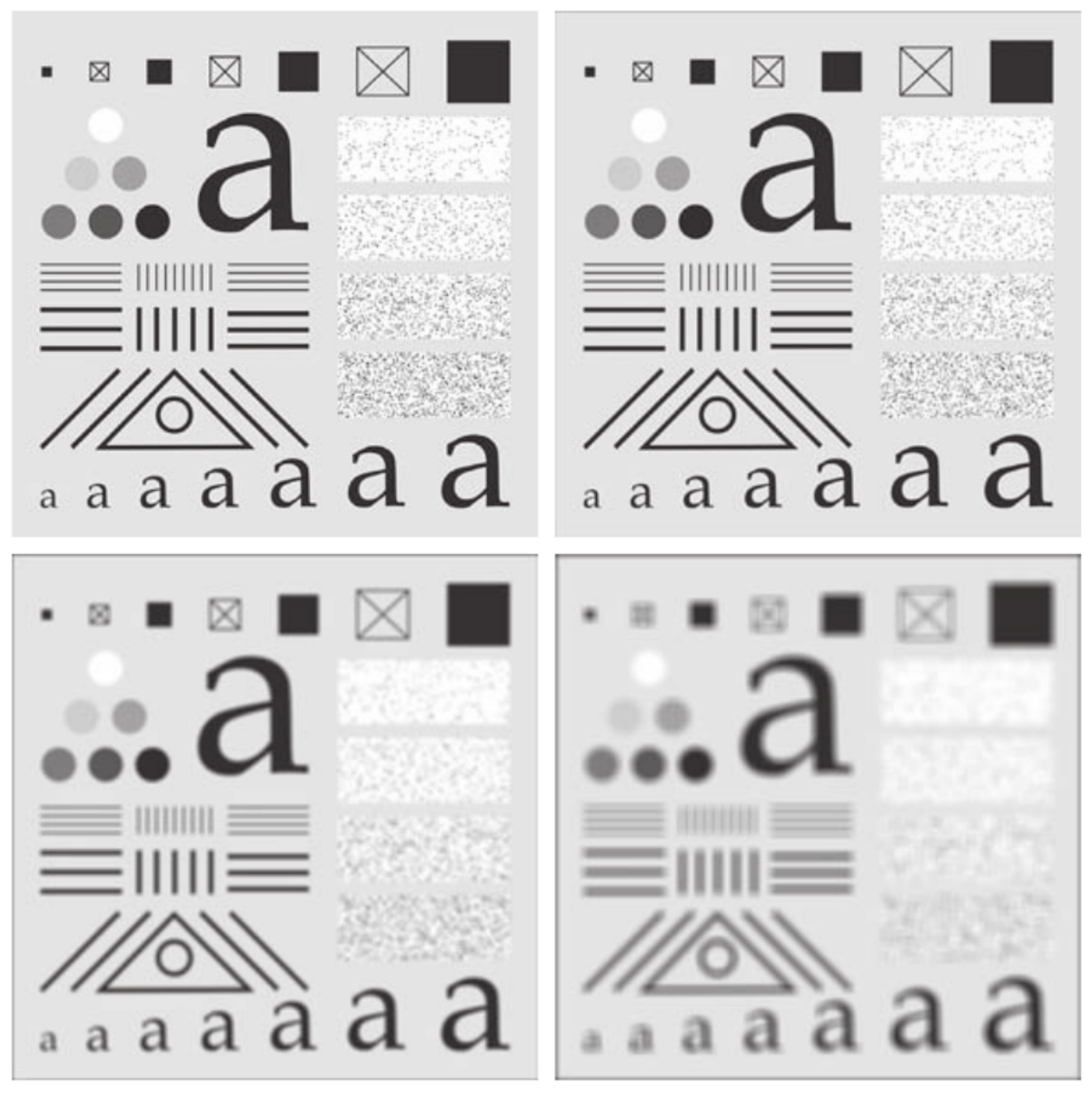
0.3679	0.6065	0.3679
0.6065	1.0000	0.6065
0.3679	0.6065	0.3679



Smoothing Spatial Filters

(a) Test pattern of size 1024×1024 pixels.

(b)-(d) Results of low pass filtering
with **box kernels** of sizes 3×3 ,
 11×11 , and 21×21 , respectively.

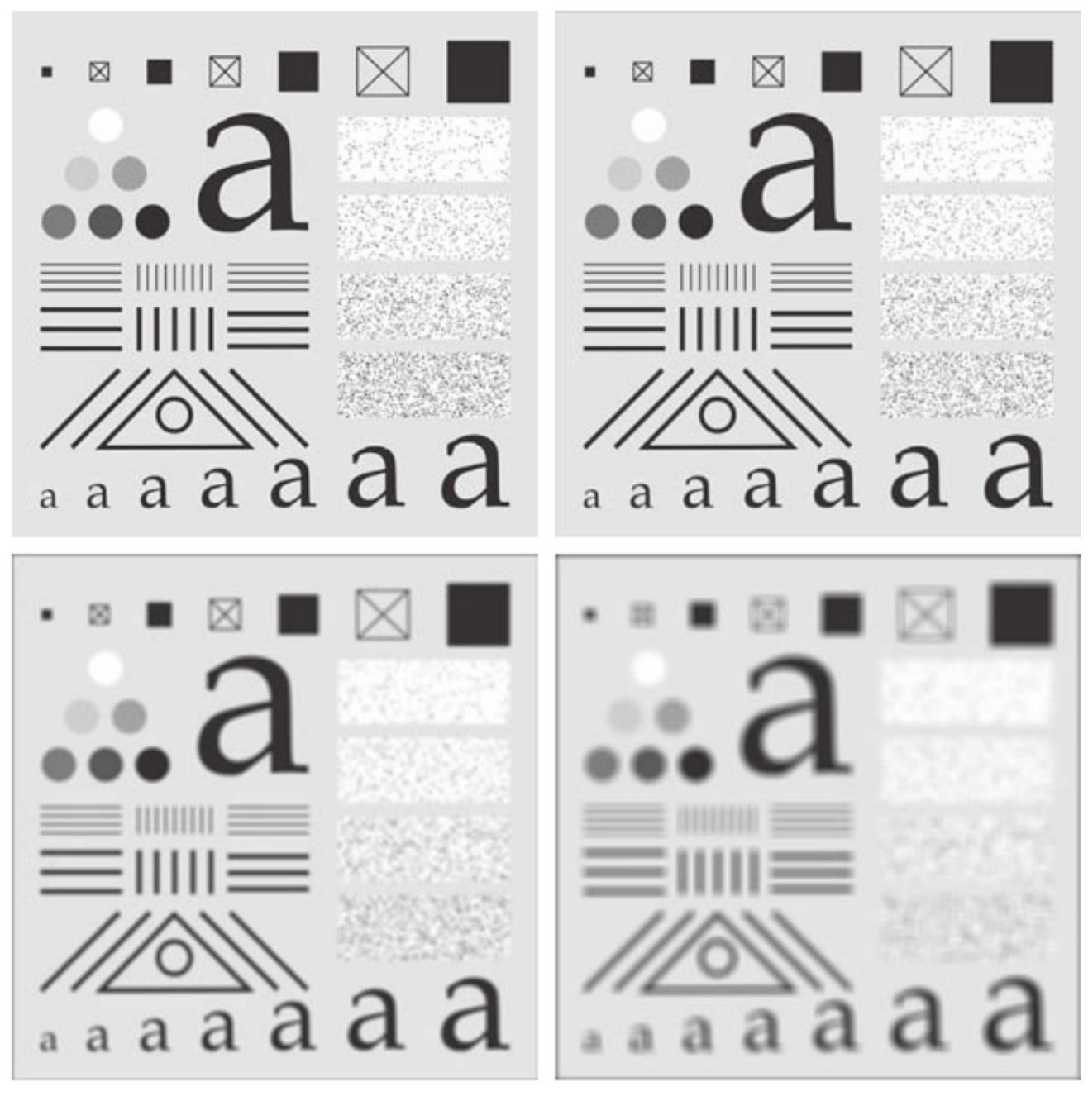




Smoothing Spatial Filters

(a) Test pattern of size 1024×1024 pixels.

(b)-(d) Results of low pass filtering
with **box kernels** of sizes 3×3 ,
 11×11 , and 21×21 , respectively.





Spatial Filtering

- A spatial filter consists of (a) **a neighborhood**, and (b) **a predefined operation**
- Linear spatial filtering of an image of size MxN with a filter of size mxn is given by the expression

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$



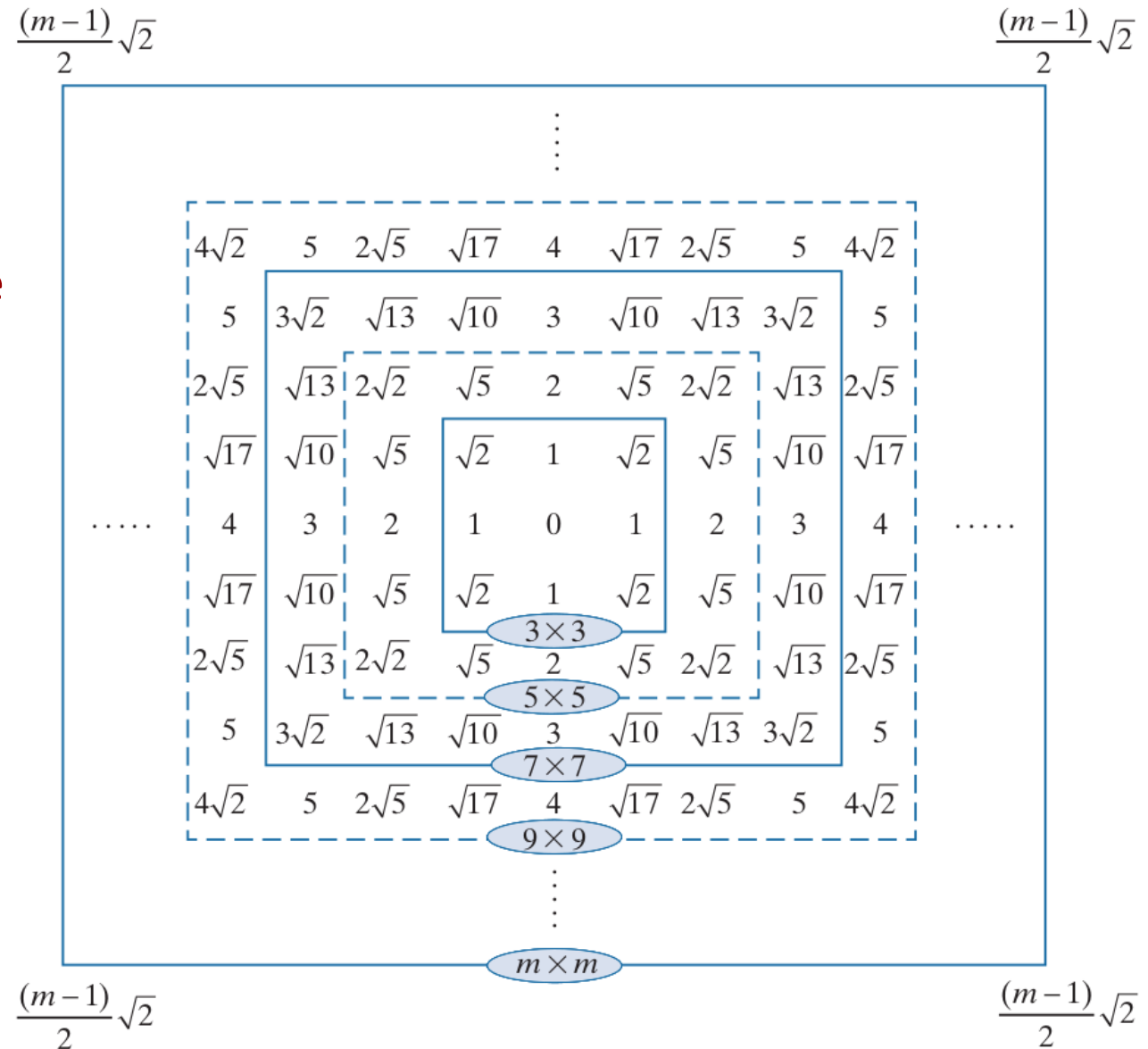
Gaussian Filters

Let s and r be position of a cell in a filter w.r.t the origin; r represent the distance of the cell from the center

$$r = [s^2 + t^2]^{1/2}$$

Form of Gaussian kernels is as below:

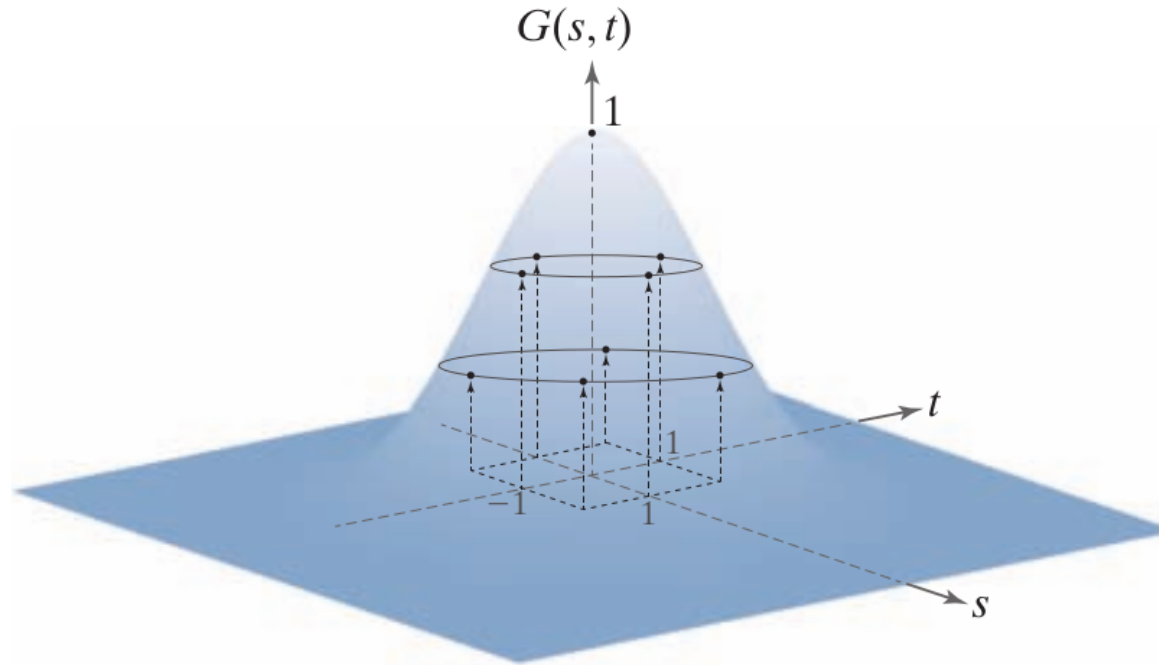
$$G(r) = Ke^{-\frac{r^2}{2\sigma^2}}$$



Distances from the center for various sizes of square kernels.



Gaussian Filters



Sampling a Gaussian function to obtain a discrete Gaussian kernel. The values shown are for $K = 1$ and $\sigma = 1$.

$$\frac{1}{4.8976} \times$$

0.3679	0.6065	0.3679
0.6065	1.0000	0.6065
0.3679	0.6065	0.3679

Resulting 3×3 kernel

$$r = [s^2 + t^2]^{1/2}$$

$$G(r) = Ke^{-\frac{r^2}{2\sigma^2}}$$



Gaussian Filters



A test pattern of size 1024×1024



lowpass filtering the pattern with a Gaussian kernel of size 21×21 , with $\sigma = 3.5$, $K=1$



Result of using a kernel of size 43×43 , with $\sigma = 7$, $K=1$



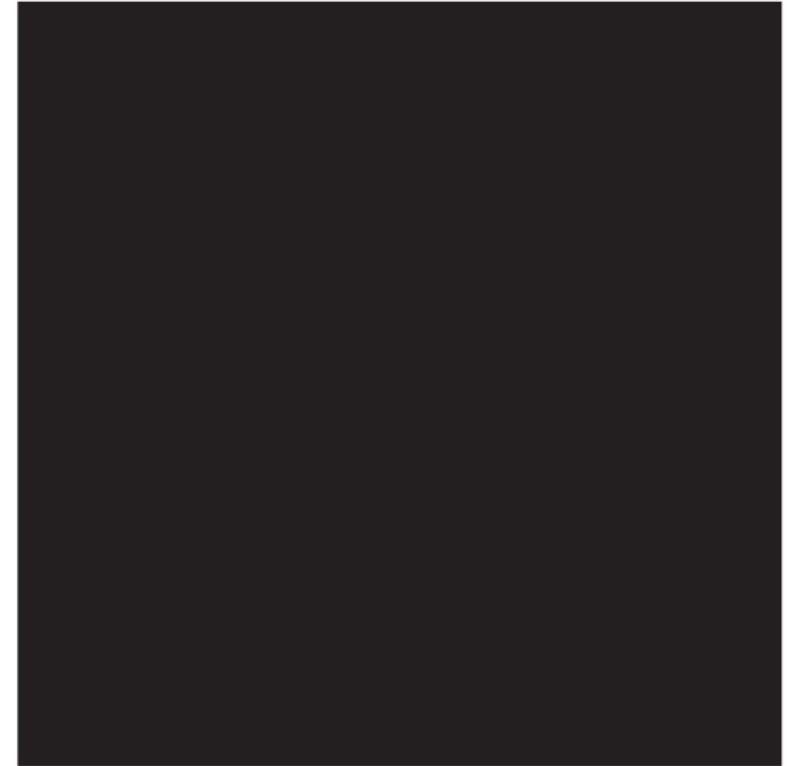
Gaussian Filters



Gaussian kernels of size 43×43 ,
with $\sigma = 7$

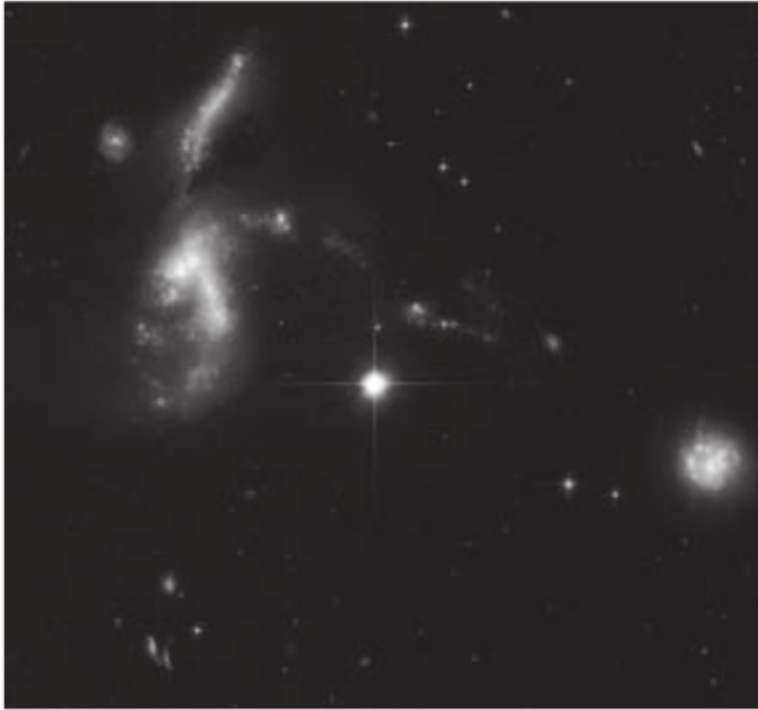


Kernel of 85×85 , with $\sigma = 7$



Difference image

Gaussian Filters - Example Usage

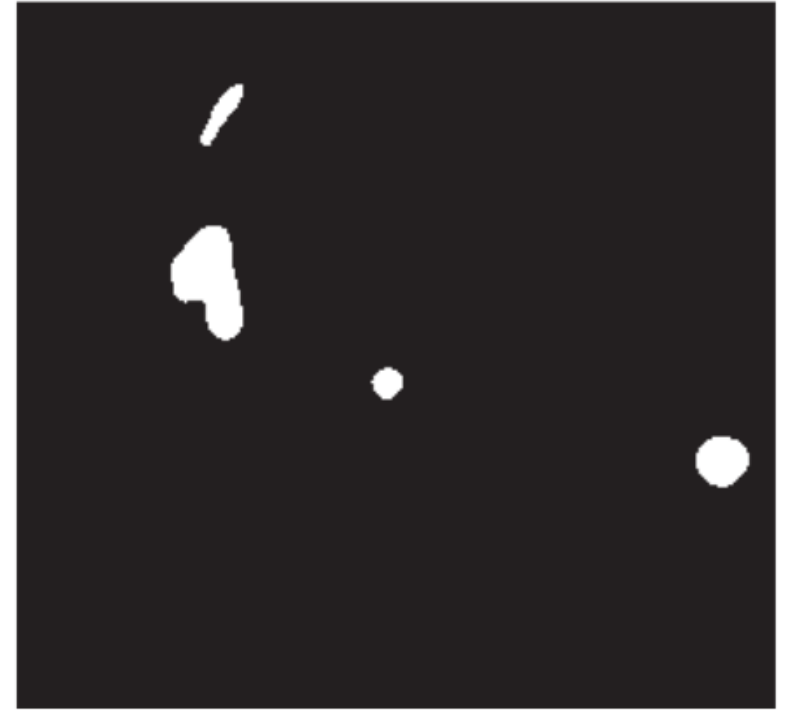


a b c

A 2566×2758 Hubble Telescope image of the Hickson Compact Group



Result of lowpass filtering with a Gaussian kernel.



Result of thresholding the filtered image (intensities were scaled to the range $[0, 1]$)



Gaussian Filters - Example Usage

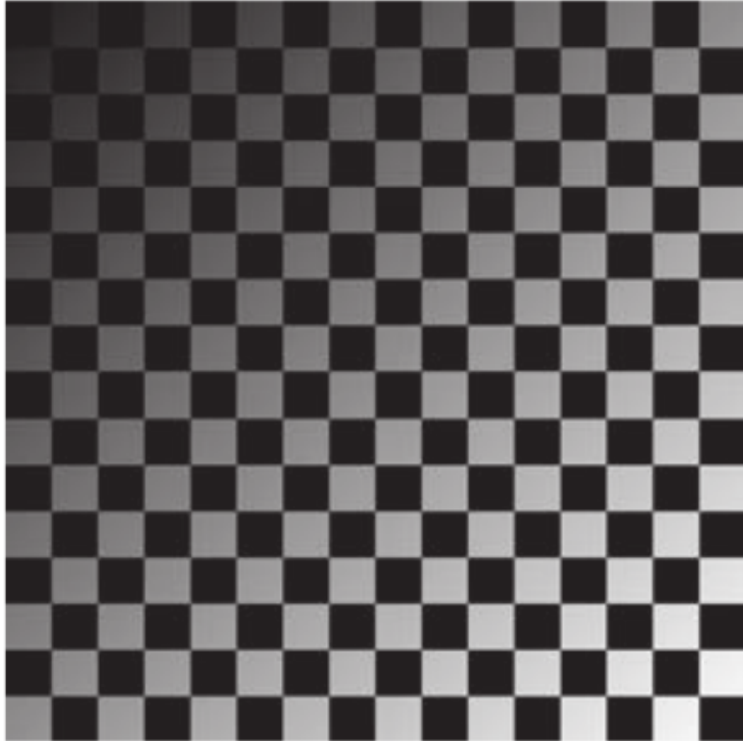
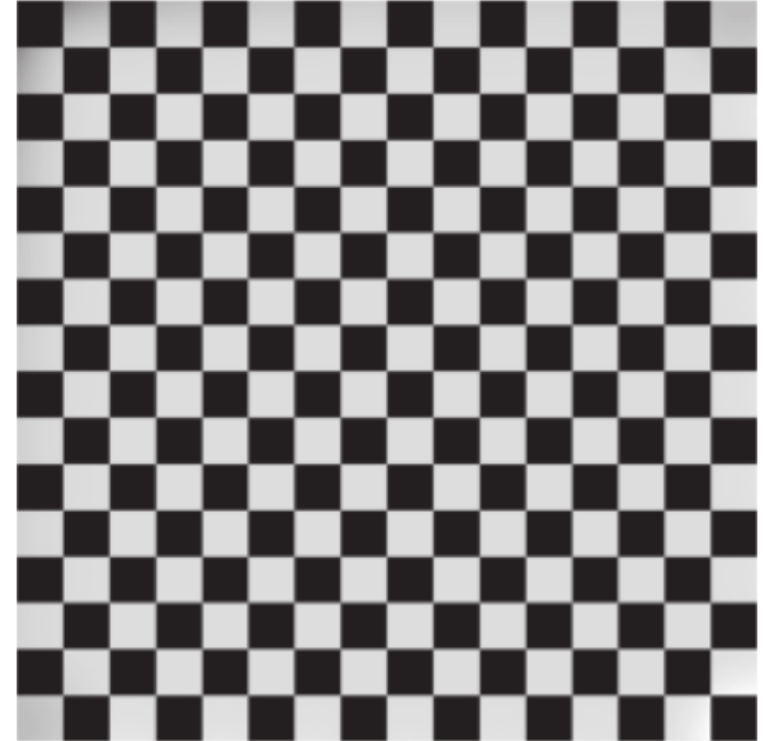


Image shaded by a shading pattern oriented in the -45° direction



Estimate of the shading patterns obtained using lowpass filtering.



Result of dividing (a) by (b). (See Section 9.8 for a morphological approach to shading correction).



- Readings:

Digital Image Processing, Rafael C. Gonzalez & Richard E woods, Third Ed, Chapter 3;

- Next Class:

Image Sharpening; Linear Filters; Edge Detection using Gradients, Sobel, Canny; Demonstration of Applications in Python

Line detection using Hough transforms; [if possible]

Thank you